

DSA Day 18: Recursion vs Iteration & tail recursion

Book: Data Structures Through C in Depth - Ch5 §5.6-5.9 · pp 166-175 | Code in Python

Overview

Iteration is usually cheaper (no call overhead). Tail recursion recurses as the last action; direct vs indirect recursion.

Key concepts to master

- Tail call
- Converting recursion $\leftrightarrow$ iteration
- Direct vs indirect

Python implementation

```
def fact_tail(n, acc=1):  
    return acc if n==0 else fact_tail(n-1, acc*n)  
def fact_iter(n):  
    r=1  
    for i in range(2, n+1): r*=i  
    return r
```

Complexity

Both $O(n)$; iteration uses $O(1)$ stack.

Common pitfalls

- Python does NOT optimise tail calls
- Assuming recursion is always elegant=better

Practice

- Compare both factorials
- When is iteration better?

